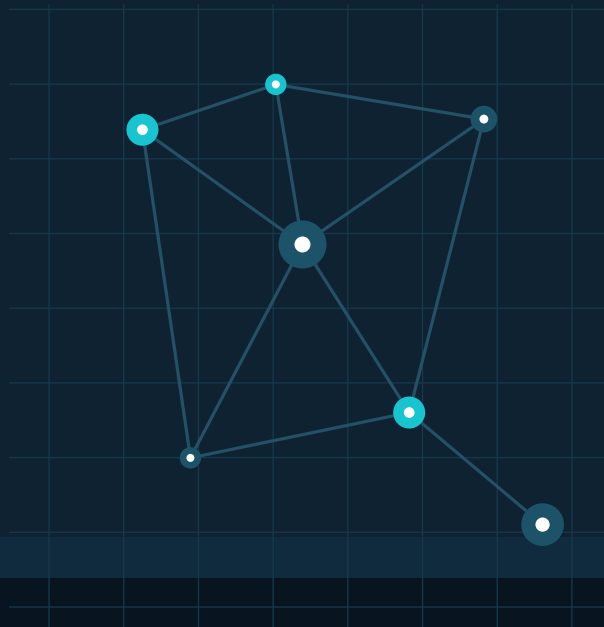


TELECOM CUSTOMER CHURN PREDICTION

Professional analytical report | Logistic Regression
Model performance • churn risk • decision threshold • business action



PROJECT OBJECTIVE

Turn customer data into an actionable churn signal.

Predict whether a customer is likely to churn, estimate churn probability, and support earlier retention action using customer, service, contract, payment and billing information.

7,043

CUSTOMERS

21

COLUMNS

80.6%

ACCURACY

0.842

ROC-AUC

BUSINESS FOCUS

Reduce missed churn without letting retention costs grow unchecked.

Baseline recall: 55.9% • 0.40 threshold recall: 66.8% • Source: supplied analysis notebook

MODEL-DRIVEN INSIGHT

01 Executive Overview

What this notebook delivers	A supervised binary classification workflow that turns telecom customer attributes into a churn probability and a stay/churn decision.
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80.6% Accuracy	65.7% Precision	55.9% Recall	0.842 ROC-AUC
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The model Logistic Regression is wrapped in a preprocessing pipeline. Numeric features are median-imputed and standardized; categorical features are most-frequently imputed and one-hot encoded. The notebook fits this workflow on the training split and evaluates it on 1,409 held-out customers.	The business lens The notebook emphasizes reducing missed churners. At the default 0.50 threshold, 165 actual churners are missed. A 0.40 threshold experiment raises recall to 66.8% while lowering precision to 56.9%.
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At-a-glance findings

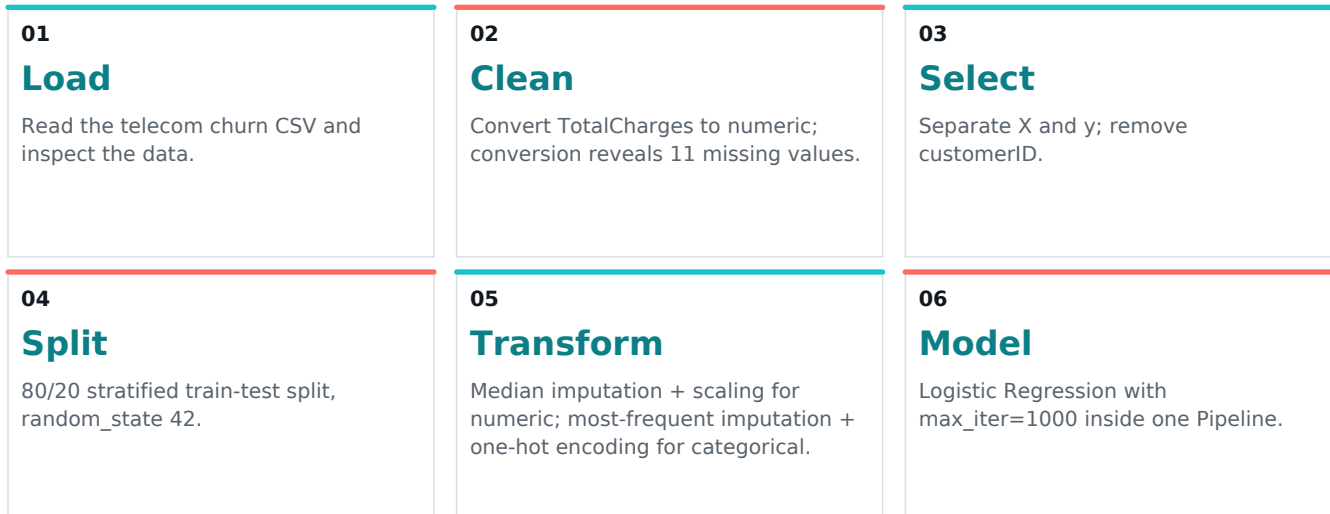
Dataset scale	7,043 records with 21 columns; 19 model input features after removing target and identifier.
Class distribution	5,174 stayed and 1,869 churned, so churn represents about 26.5% of the dataset.
Test-set quality	Classification report shows stronger performance on Stay (precision 0.85, recall 0.89) than Churn (precision 0.66, recall 0.56).
Operational focus	The notebook recommends focusing on false negatives and testing a lower decision threshold.

02 Data Foundation & Preparation

Dataset profile

Rows	7,043	Columns	21
Target	Churn (No / Yes)	Model inputs	19 after dropping customerID and Churn
Training split	5,634 records	Test split	1,409 records

Preparation workflow

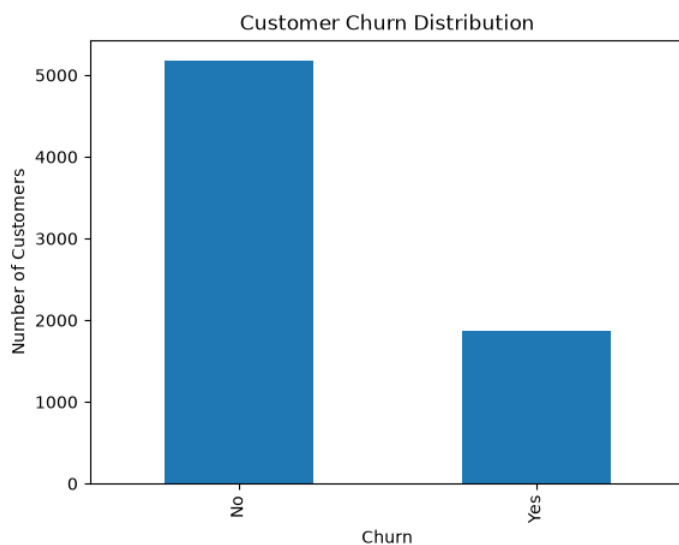


Data-quality note The notebook initially reports zero nulls, but TotalCharges is stored as text. After numeric conversion with errors="coerce", 11 TotalCharges entries become missing and are then handled by the numeric median-imputation step in the preprocessing pipeline.

Input feature groups

Numerical: SeniorCitizen, tenure, MonthlyCharges, TotalCharges. Categorical: gender, Partner, Dependents, PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies, Contract, PaperlessBilling and PaymentMethod.

03 Customer Churn Landscape



Class balance

The target contains 5,174 customers who stayed and 1,869 who churned. That means the positive churn class is the minority, at approximately 26.5%.

Why it matters

A model that simply favors the larger Stay class can still look accurate while missing churners. The notebook therefore reports precision, recall, F1-score and ROC-AUC alongside accuracy.

Analytical takeaway

The objective is not only to classify correctly overall, but to identify enough true churn cases to support proactive retention.

Descriptive statistics from the notebook

Feature	Mean / central value	Observed range
SeniorCitizen	0.162	0 to 1
tenure	32.37 months	0 to 72 months
MonthlyCharges	64.76	18.25 to 118.75
TotalCharges	Not summarized in the notebook output	Converted to numeric before modeling

Scope note

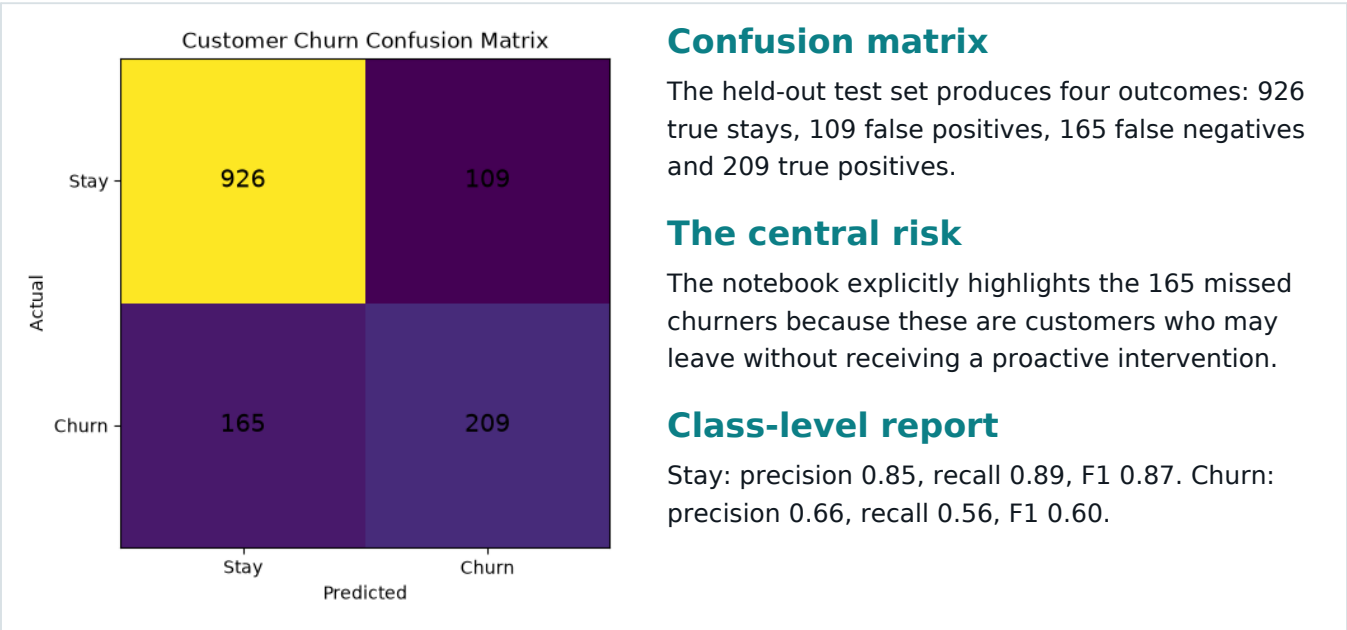
This notebook focuses its visual EDA on overall churn distribution. No additional group-level churn-rate charts are present in the supplied notebook, so this report does not invent category-level driver findings.

04 Model Performance & Error Profile

Logistic Regression

The notebook uses a Pipeline that combines preprocessing and Logistic Regression. This keeps transformation steps attached to the model and lets the workflow generate both class predictions and churn probabilities.

Metric	Value	Interpretation
Accuracy	0.806	80.6% of held-out customers were classified correctly.
Precision	0.657	65.7% of customers flagged as churn were actually churners.
Recall	0.559	55.9% of actual churners were captured.
F1-score	0.604	Balance of precision and recall for the churn class.
ROC-AUC	0.842	Strong ranking ability based on the notebook output.



209

churners correctly identified

165

churners missed

109

stayers incorrectly flagged

05 Decision Threshold & Unseen Customers

Default classification rule

The notebook uses the model prediction at its default 0.50 decision boundary, producing the baseline results above. It then tests a lower 0.40 threshold to favor recall for churn.

Decision rule	Accuracy	Precision	Recall	F1
0.50 baseline	0.806	0.657	0.559	0.604
0.40 experiment	0.778	0.569	0.668	0.615

RECALL +10.9 pts	Precision -8.8 pts	Accuracy -2.8 pts	F1 +1.1 pts
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Unseen customer prediction

5%	Likely to Stay	95%
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Prediction	Likely to Stay
Churn probability	0.05 (5%)
Notebook basis	The first held-out customer, X_test.iloc[[0]], was passed through the fitted model after training.
First 10 held-out records	Probabilities range from 0.002881 to 0.684411; two of the ten are predicted as likely to churn (0.684411 and 0.603935).

06 Business Interpretation & Recommendations

REDUCE MISSED CHURN Focus on the 165 false negatives. These are actual churners that the baseline classifier did not flag.	CONTROL OFFER WASTE 109 false positives mean some stayers receive an unnecessary retention signal. A lower threshold increases this trade-off.	VALIDATE BEFORE DEPLOY The 0.40 experiment improves recall in this notebook, but the final cutoff should be selected using offer cost and churn-loss economics.
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Recommended action framework

Risk signal	Suggested business response	Reason
High churn probability	Prioritize proactive retention outreach.	The model is explicitly intended to identify customers who may churn early.
Borderline probability	Consider a lighter-touch intervention or monitor.	A lower threshold catches more churners but also creates more false positives.
Low churn probability	Use standard service / loyalty engagement.	The unseen example at 5% was predicted as likely to stay.

Conclusion

The supplied notebook demonstrates a complete churn-classification pipeline: data loading, data conversion and imputation, one-hot encoding, scaling, stratified splitting, Logistic Regression, probability scoring, classification, confusion-matrix analysis and threshold experimentation. The baseline model reaches 80.6% accuracy and 0.842 ROC-AUC, while the 0.40 threshold experiment increases churn recall from 55.9% to 66.8%. The clearest business priority is to reduce missed churners without letting retention costs grow unchecked.

Source basis: customer_churn_analysis.ipynb supplied in this conversation. All reported metrics, counts, model steps and predictions are taken from that notebook output.